

Integrating Network Function Virtualization with the DevOps Pipeline: Kubernetes

The third part of this series of articles on integration of network function virtualization with the DevOps pipeline discusses the architecture of Kubernetes and the various ways to build a cluster for your infrastructure.



Kubernetes is open source container orchestration software. With Kubernetes, users can run multiple containers over many different machines and automate the life cycle of containers over such distributed systems. Kubernetes can automate application deployment, scale the size of the application, and manage the containers. It can scale the application using automation without increasing the size of the operations team. In short, it is container-centric management software.

Kubernetes is often known as the kernel of distributed systems because it abstracts the mounting hardware of the nodes away from the application that is running. It provides a common ground for the workload deployed and the applications running to consume the shared pool of resources. This handling of resources by Kubernetes eases the life cycle of deployment of a large scale containerized application.

Some of the features of Kubernetes include:

- Automated rollouts and rollbacks

- Service discovery and load balancing
- Storage orchestration
- Secret and configuration management
- Horizontal scaling
- Self-healing

Architecture

When we talk about Kubernetes architecture, there are two components to look out for — master and worker nodes. These nodes follow the master/slave architecture. The master node